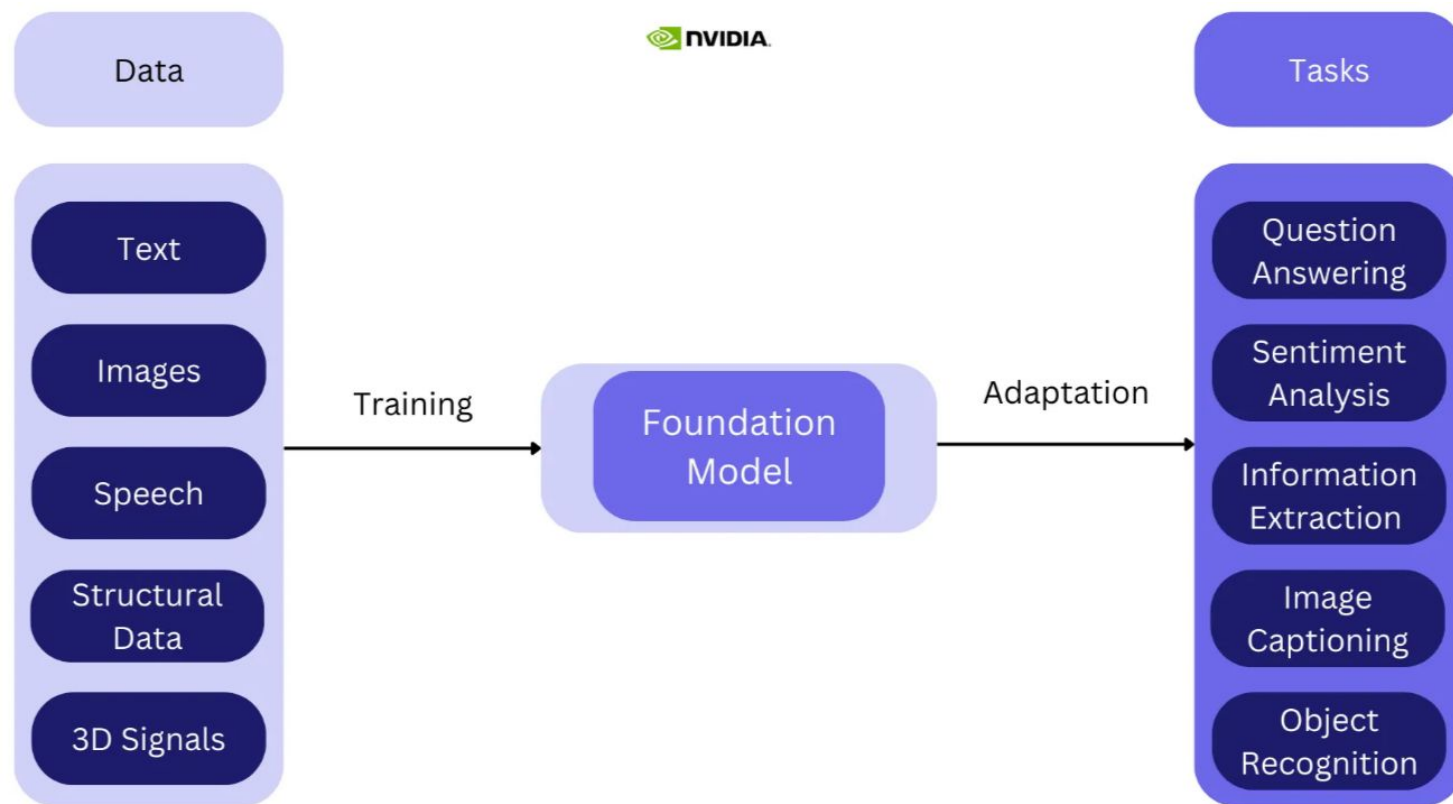


Foundation Model I

Guosheng Hu

Foundation Model

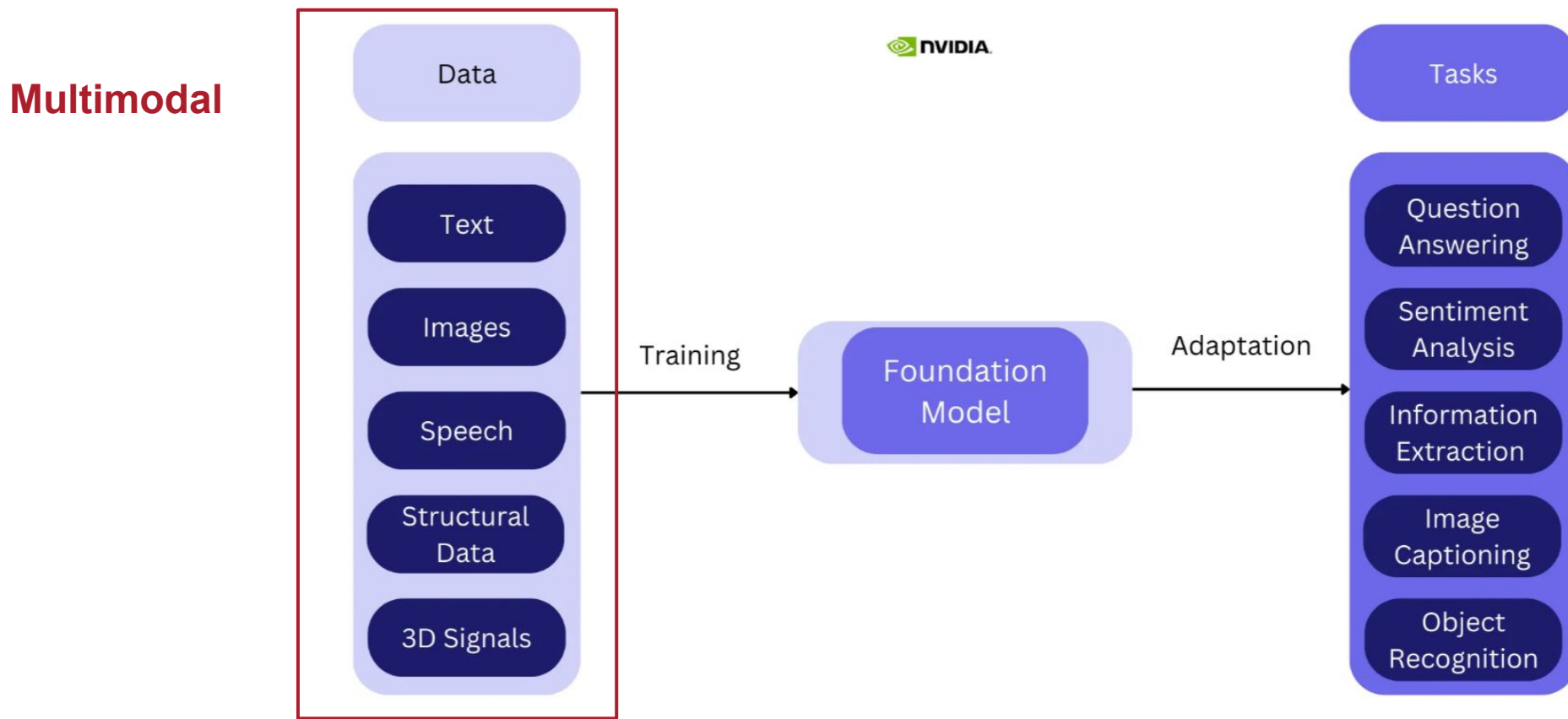
A **foundation model** is a **large, pre-trained model** (usually trained on vast and diverse datasets) that can be **adapted (fine-tuned or prompted)** for a wide range of downstream tasks.



https://medium.com/@tenyks_blogger/the-foundation-models-reshaping-computer-vision-b299a91527fb

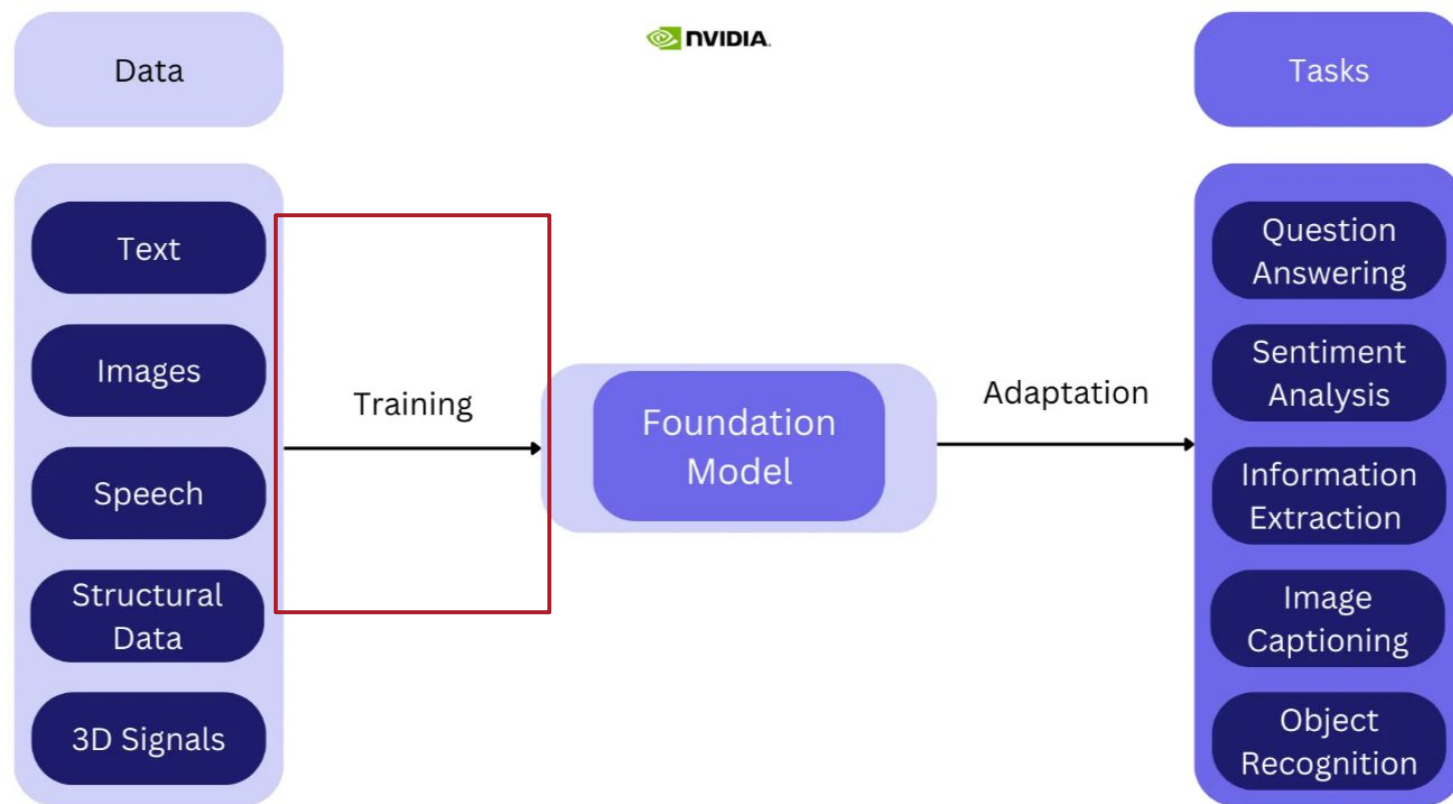
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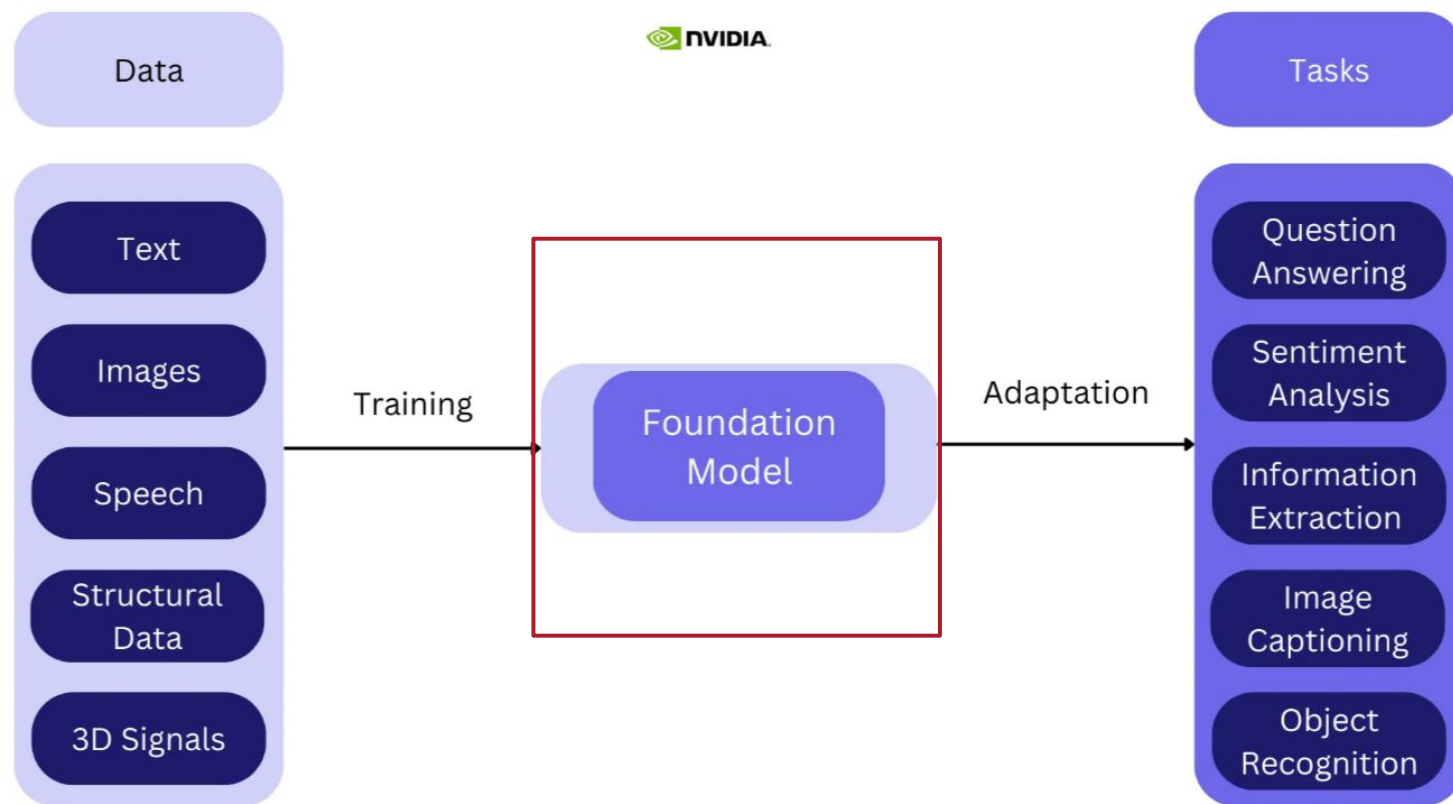
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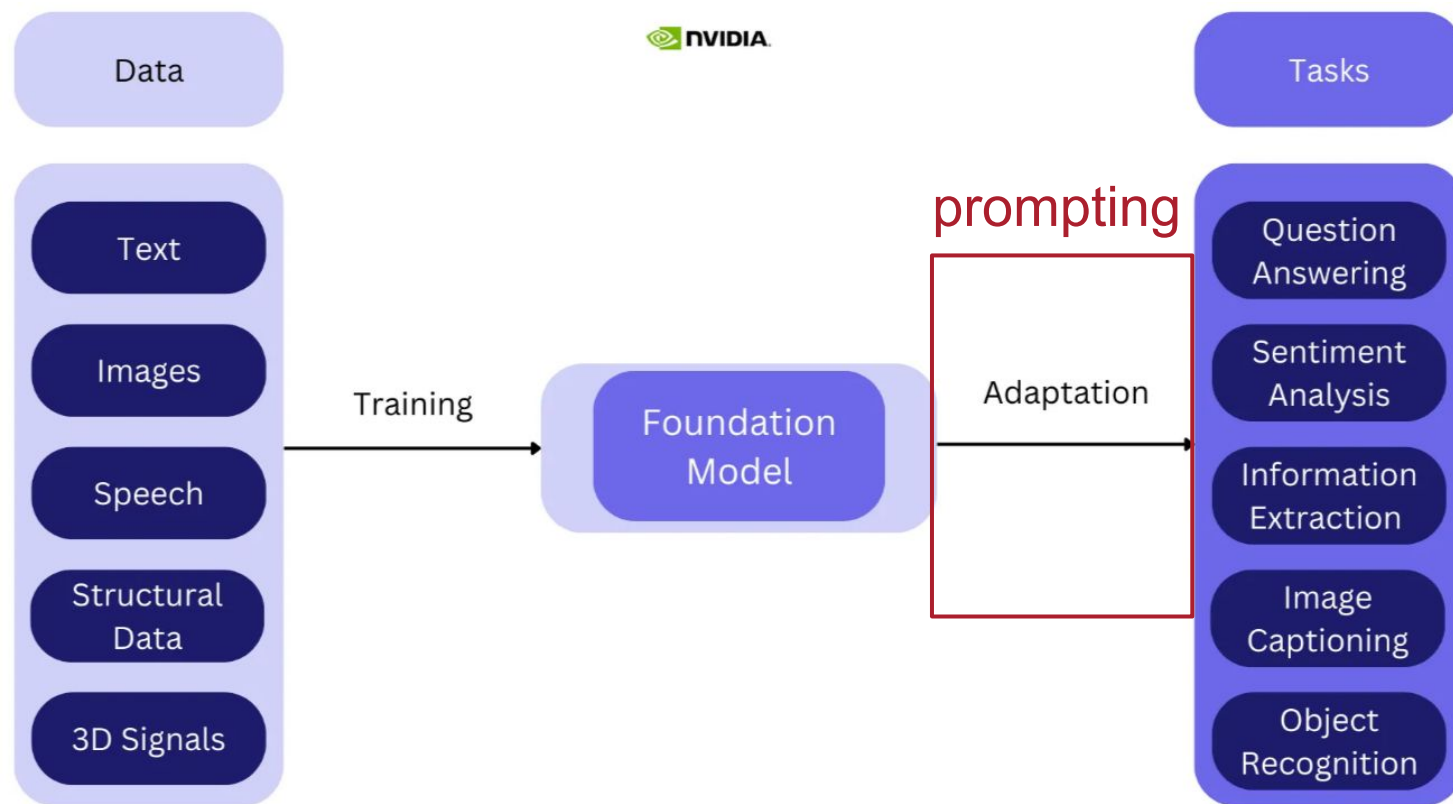
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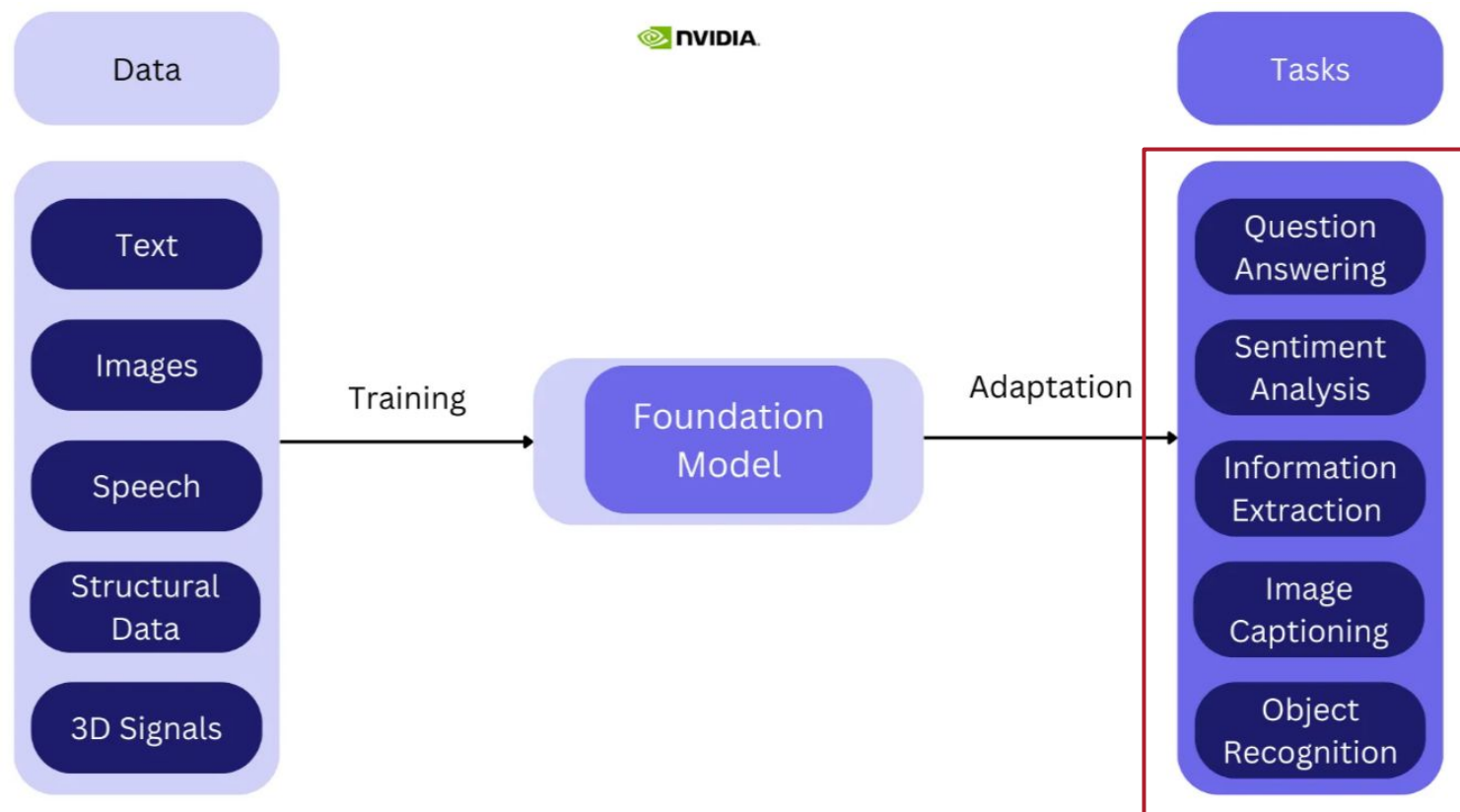
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Domain	Foundation Model	Description
Language	GPT (OpenAI), LLaMA (Meta), Claude (Anthropic)	Trained on large text corpora
Vision	DINO, SAM (Meta)	Trained on images
Vision–Language	CLIP, LLaVA, Stable Diffusion	Combine visual and textual understanding
Biology	AlphaFold, ESM	Trained on protein sequences and structures

Foundation Model

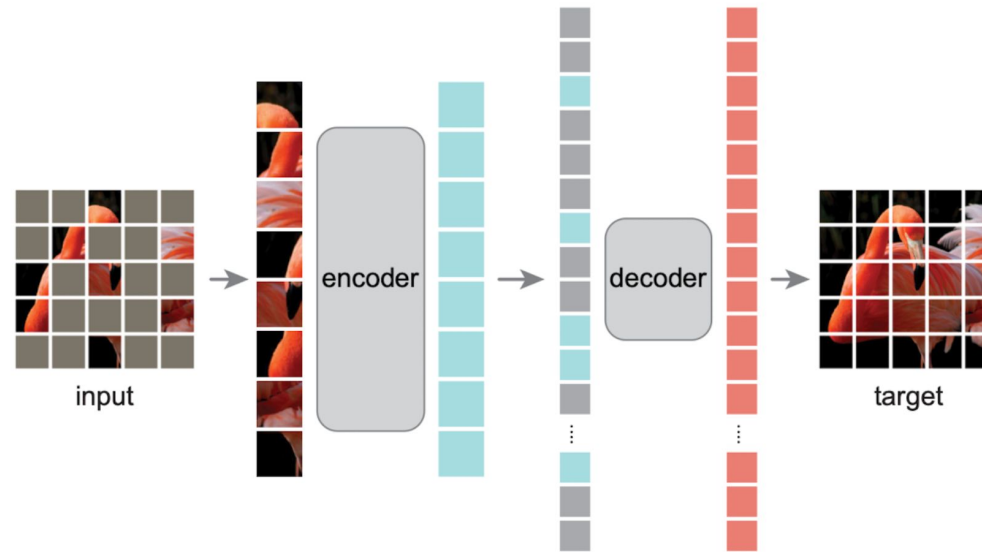
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Self-supervised Learning

We have a large number of images on Internet. It is very expensive the annotate them for training.
Can we train a very powerful model using these images without labels?

Yes, Self-Supervised learning.

Self-supervised learning for vision models trains systems to learn meaningful visual representations by solving automatically generated tasks from unlabeled image data, for example, Masked AutoEncoder and DINO (self-Distillation with NO labels).



He, Kaiming, et al. "Masked autoencoders are scalable vision learners." *CVPR*, 2022.

DINO - Data

Source dataset is **ImageNet**, a **large-scale** visual database containing over **14 million** images spanning **20,000+ object** categories.

The images are then **augmented** by **color jittering**, **Gaussian blur**, **solarization**, and **multi-crop**.



1. Original Image



2. Color Jittering



3. Gaussian Blur



4. Solarization



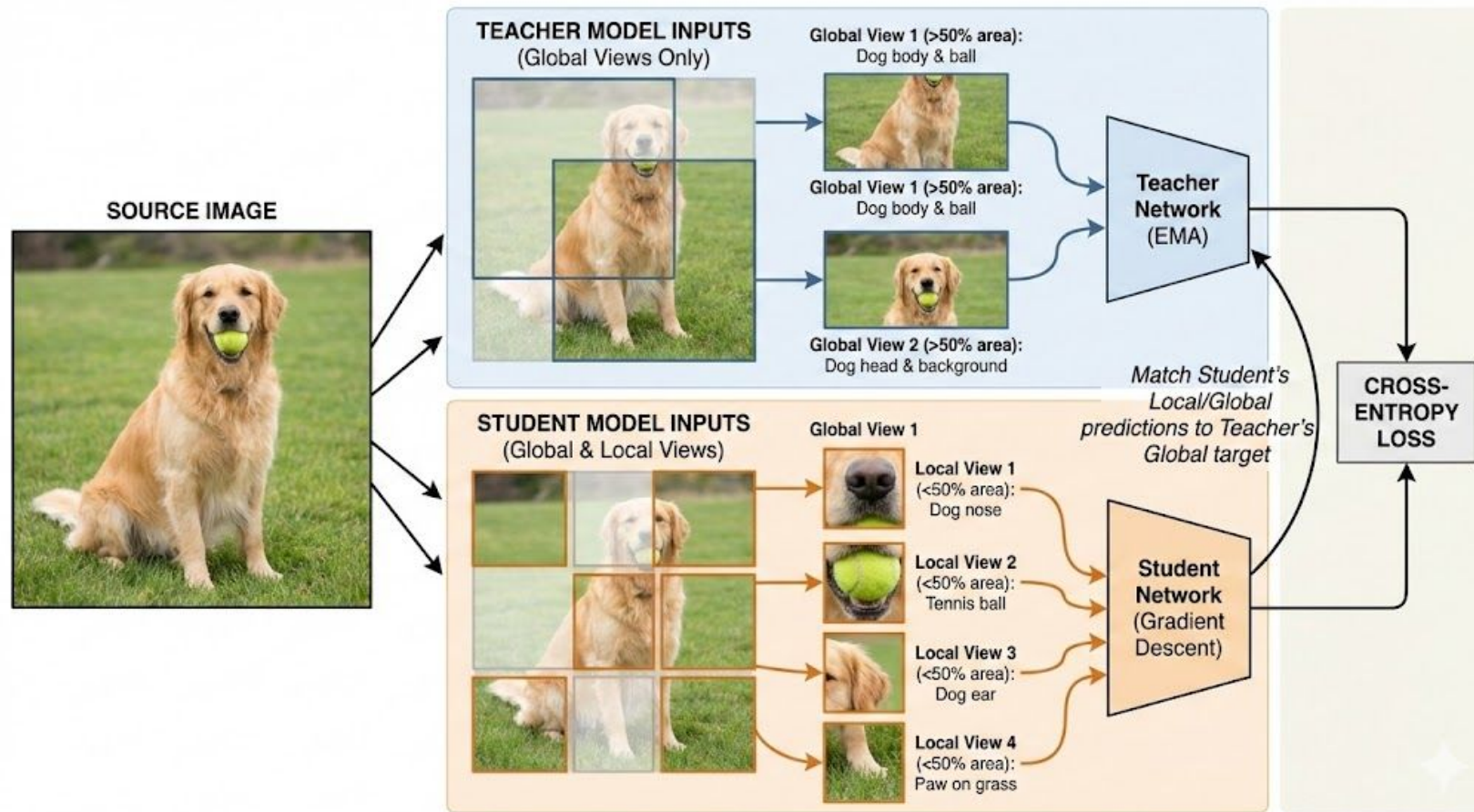
5. Multi-crop (Global)



6. Multi-crop (Local)



DINO - Training (Arch. and Loss)



For sample pairs (e.g. 2x4 in left image):

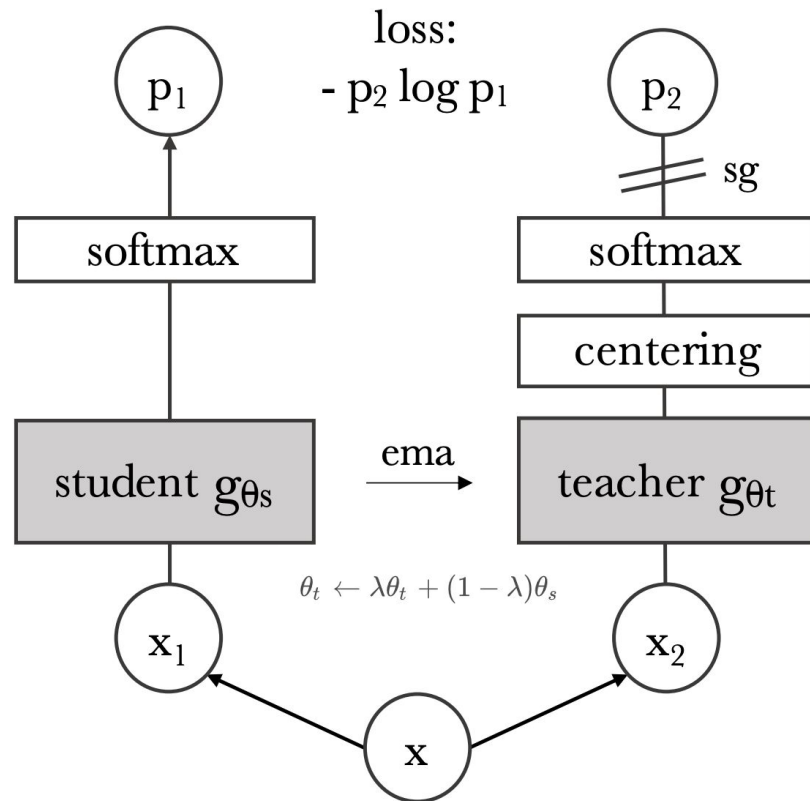
$$P_s(x)^{(i)} = \frac{\exp\left(g_{\theta_s}(x)^{(i)} / \tau_s\right)}{\sum_{k=1}^K \exp\left(g_{\theta_s}(x)^{(k)} / \tau_s\right)},$$

The entropy loss help the model to correlate the local views with the global concept (a dog in this example).

Teacher & student model share the same architecture while teacher parameters are EMA results of student parameters

Caron, Mathilde, et al. "Emerging properties in self-supervised vision transformers.", ICCV. 2021.

DINO - Training (Arch. and Loss)



The student network learn to match the extracted feature from local views with that of global feature from teacher network

$$\min_{\theta_s} \sum_{x \in \{x_1^g, x_2^g\}} \sum_{\substack{x' \in V \\ x' \neq x}} H(P_t(x), P_s(x')) .$$

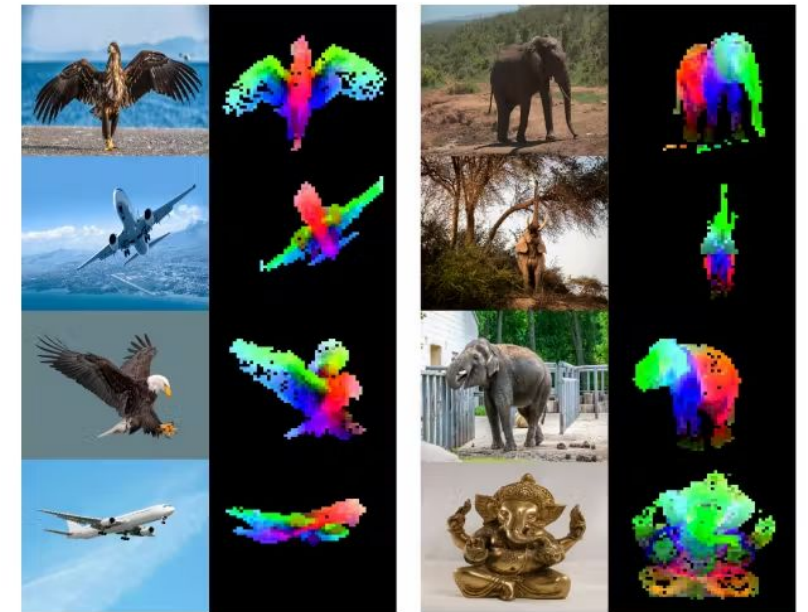
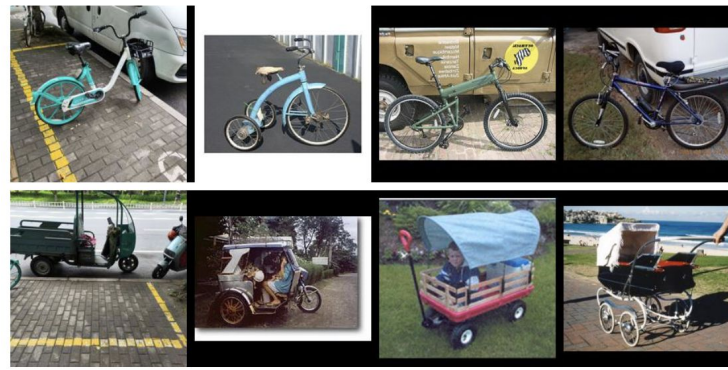
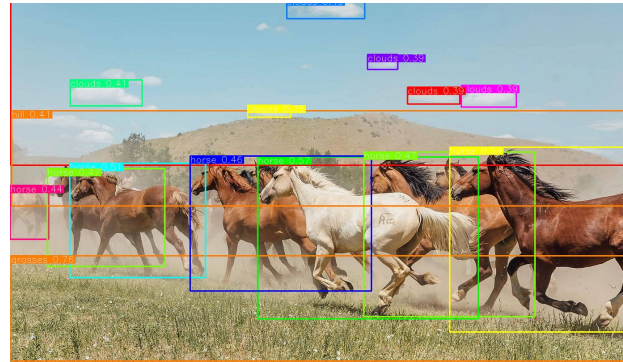
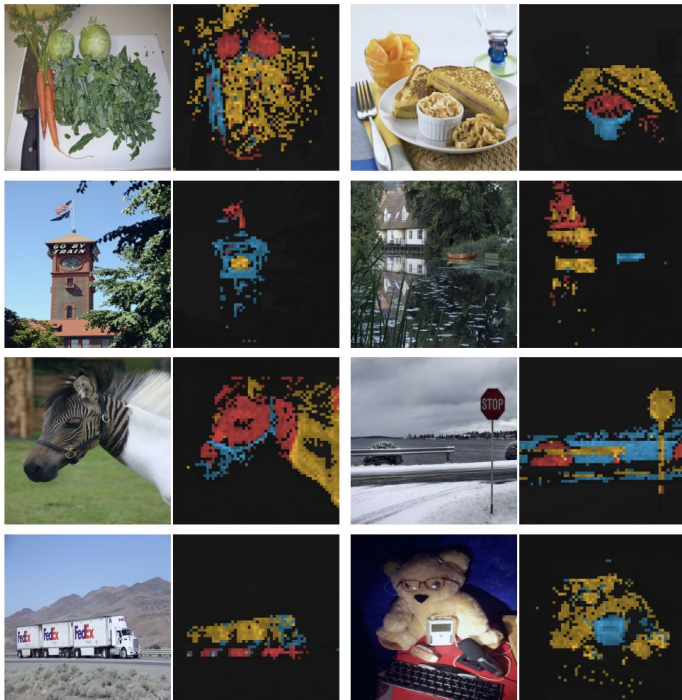
Teacher network features are modified by **centering** and **sharpening** on softmax to avoid collapse.

$$t = \text{softmax}((t - c) / tpt, \text{dim}=1)$$

$$c \leftarrow mc + (1 - m) \frac{1}{B} \sum_{i=1}^B g_{\theta_t}(x_i),$$

DINO - Applications

The DINO model can learn high-quality **image features** which are crucial for downstream visual tasks like **image retrieval**, **object detection**, and **semantic segmentation (backbone)**.

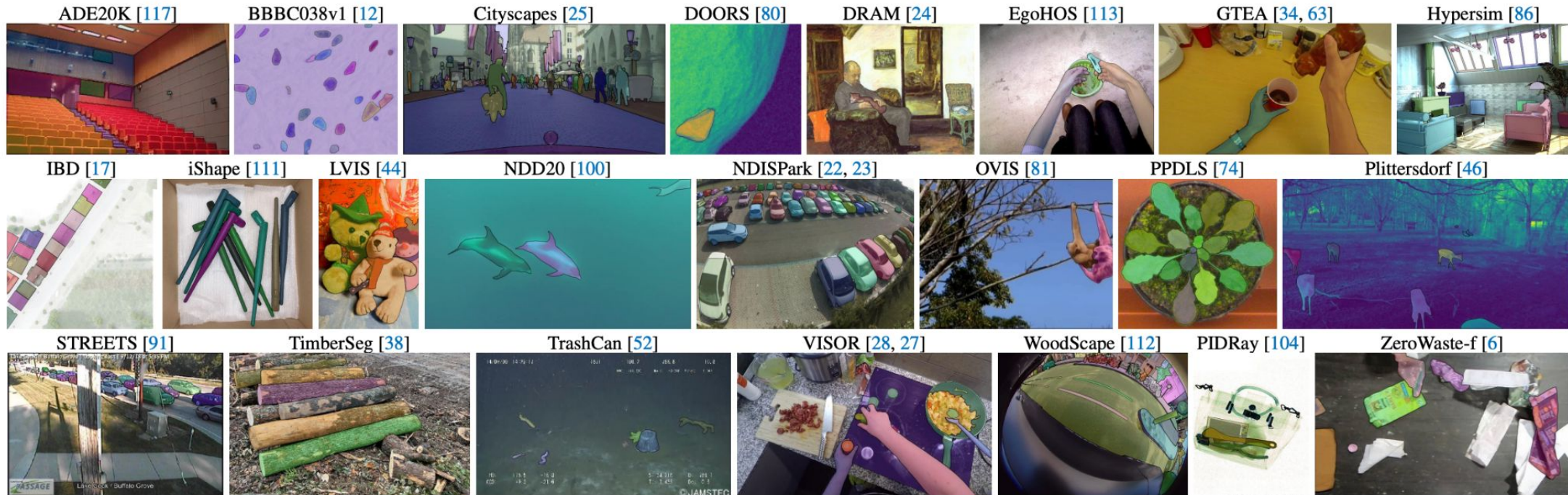


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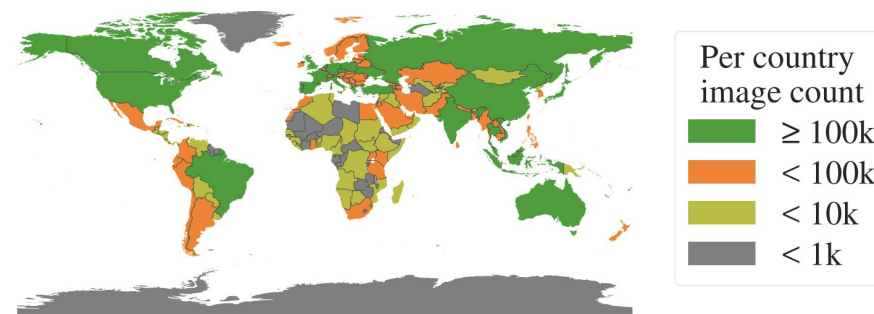
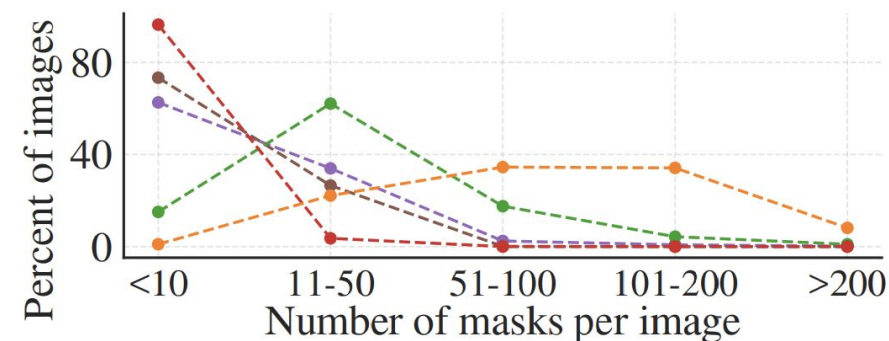
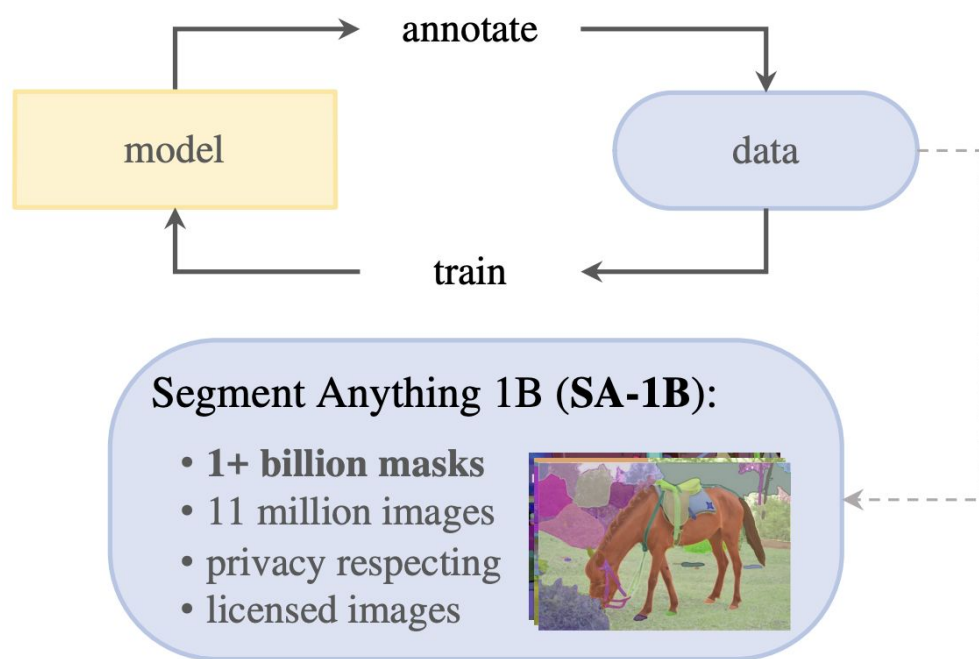
SAM - Introduction

Segment Anything Model (SAM) is a foundational model from Meta AI designed for **universal image segmentation**. It can generate a **precise mask for any object** in an image based on various **prompts**, like clicks or boxes.



SAM - Training Data

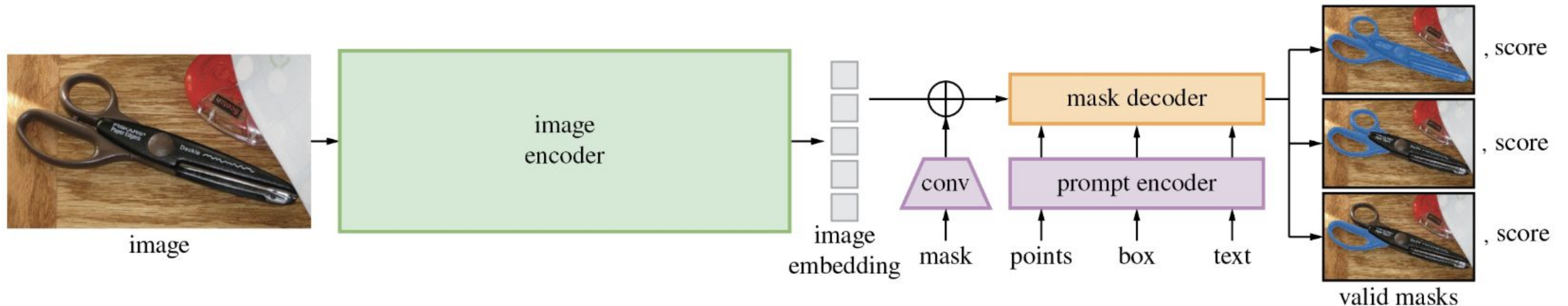
The model was trained on the massive **SA-1B dataset**, which contains over **1.1 billion high-quality segmentation masks**. This dataset was built using 11 million **diverse, high-resolution**, and **privacy-respecting** images, making it the largest of its kind.



Semi-auto annotation system is key.

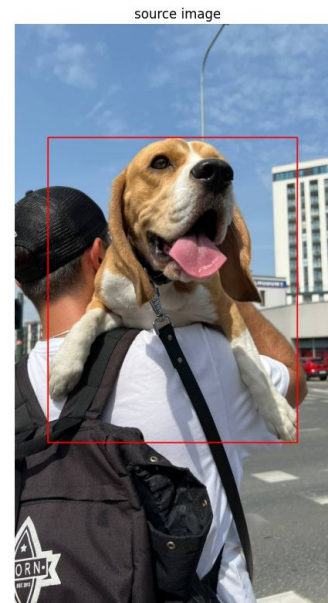
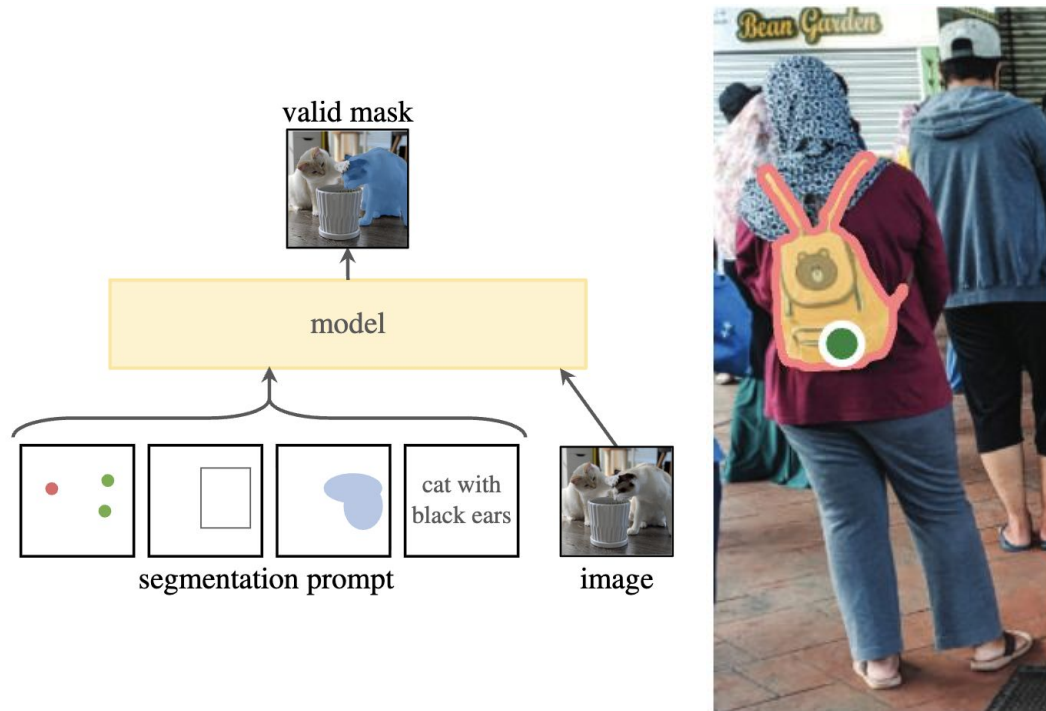
SAM - Training (Arch. and Loss)

The architecture uses a powerful **ViT-based image encoder**, a flexible **prompt encoder** (for points, boxes, etc.), and a fast, **lightweight mask decoder**. It was trained using a segmentation loss (a linear combination of Focal Loss and Dice Loss) to effectively manage class imbalance and optimise mask quality.



Kirillov, Alexander, et al. "Segment anything." *Proceedings of the IEEE/CVF international conference on computer vision*. 2023.

Prompts for SAM (Examples)



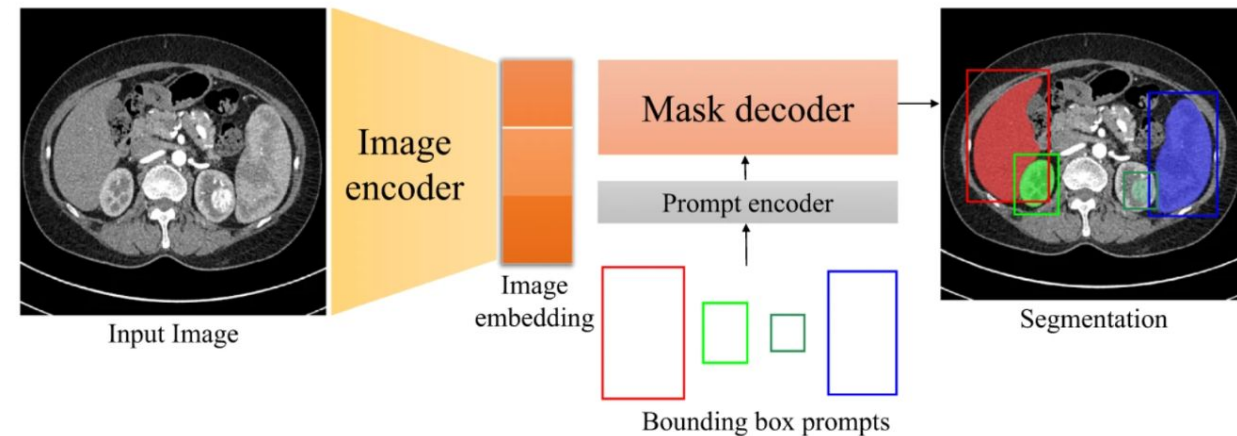
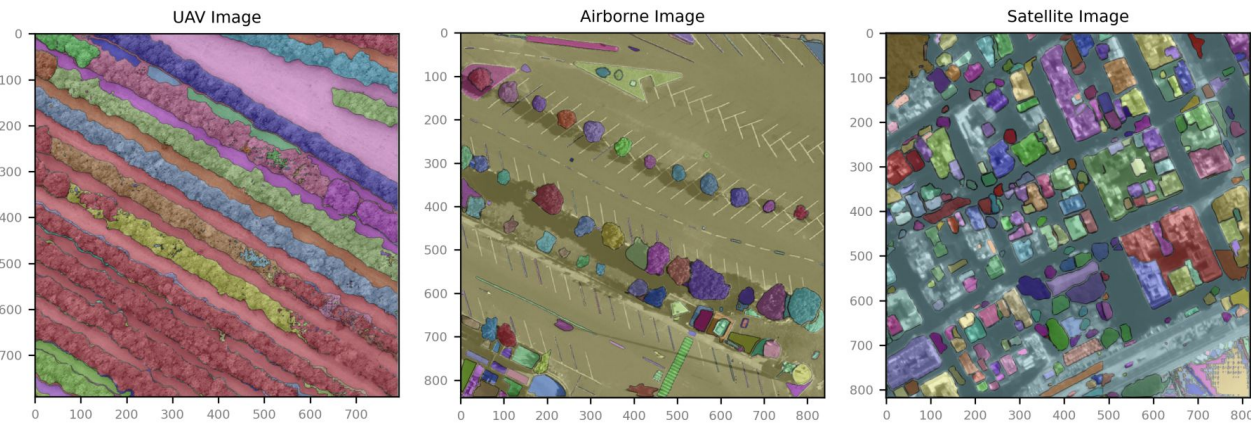
“elephant”



Kirillov, Alexander, et al. "Segment anything." *Proceedings of the IEEE/CVF international conference on computer vision*. 2023.

SAM - Applications

SAM's powerful **zero-shot capabilities** enable a wide variety of downstream applications, including creative **image editing**, **robotics**, and **scientific analysis**. For instance, it is highly effective in specialised fields like **remote sensing** for identifying land features and in **medical imaging** for analysing complex anatomical structures.



SAM - Challenges and Opportunities

- **Efficiency**

- SAM is large; compressing or distilling it for edge devices is an open research problem.

- **3D segmentation**

- Can we adapt SAM (for 2D images) to 3D segmentation?

- **Security and privacy**

- Traditional segmentation models output fixed number of classes, SAM not. It is not good for healthcare image segmentation. e.g. prompt 'segment abnormalities', it might release patients' privacy like hidden pathologies and previously unreported findings.

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AlphaFold - Background

2024 Nobel Prize Winner in Chemistry

Proteins can be represented as Amino Acid Sequence like: Met–Leu–Gly–... (M–L–G–...)

We can easily get protein/Amino Acid sequence

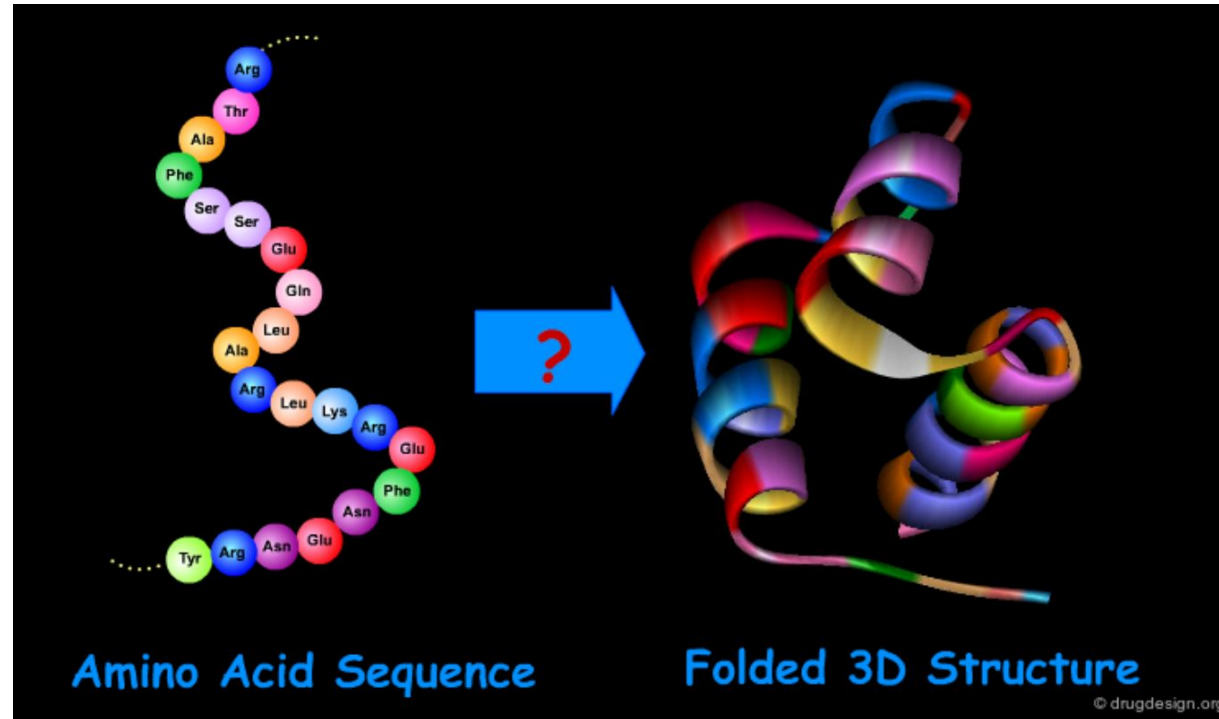
But predicting 3D structure from sequence was extremely hard/expensive.

But Protein function depends on **3D shape**

Known as the “**protein folding problem**”

Jumper, John, et al. "Highly accurate protein structure prediction with AlphaFold." *nature* 596.7873 (2021): 583-589.

Learning Problem



$$\text{M-L-G-...} \xrightarrow{f(.)} (x_1, y_1, z_1, x_2, y_2, z_2, \dots)$$

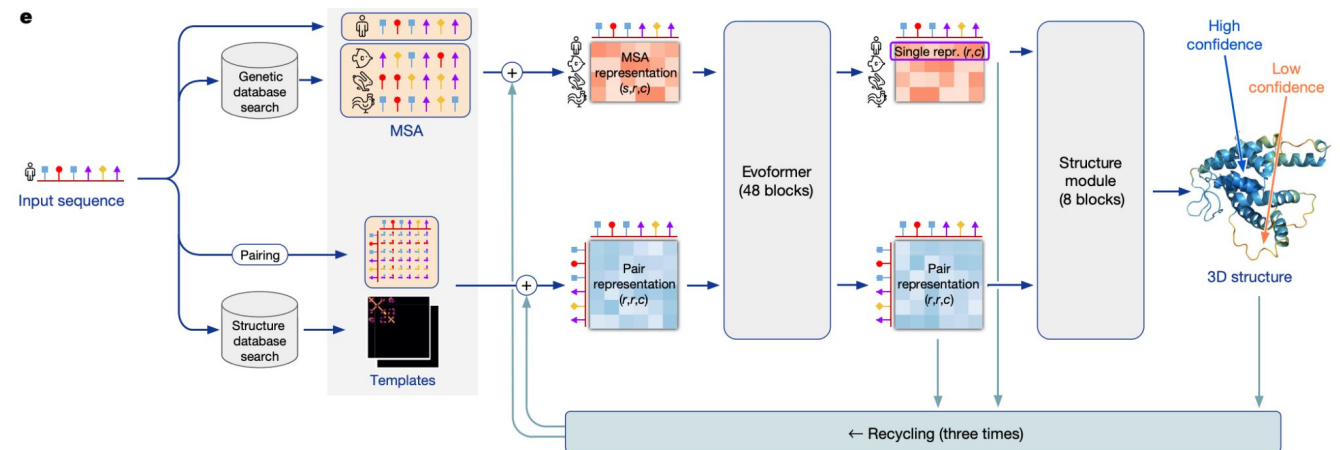
<https://www.drugdesign.org/chapters/protein-structure/#link-between-protein-sequence-folding-and-function>

AlphaFold

$$\text{M-L-G-...} \xrightarrow{f(.)} (x_1, y_1, z_1, x_2, y_2, z_2, \dots)$$

$$f(\text{sequence} + \text{evolutionary information}) \rightarrow \text{3D atomic coordinates}$$

Evolution leaves statistical footprints. If two positions change together across species, they are probably close in 3D. AlphaFold looks at many related protein sequences (constructed using Multiple Sequence Alignment, MSA), it detects patterns like which positions mutate together.



Jumper, John, et al. "Highly accurate protein structure prediction with AlphaFold." *nature* 596.7873 (2021): 583-589.

AlphaFold

- Before AlphaFold: ~200,000 protein structures were experimentally known.
- After AlphaFold: Structures predicted for ~200 million proteins.
- Accelerating Drug Discovery
- Advancing Biology Globally
 - AlphaFold didn't just improve protein structure prediction, it changed how biology research is done.

Jumper, John, et al. "Highly accurate protein structure prediction with AlphaFold." *nature* 596.7873 (2021): 583-589.

Reading List

He, Kaiming, et al. "Masked autoencoders are scalable vision learners." *CVPR*, 2022.

Kirillov, Alexander, et al. "Segment anything." *Proceedings of the IEEE/CVF international conference on computer vision*. 2023.

Oquab M, Darcet T, Moutakanni T, et al. "DINOv2: Learning robust visual features without supervision." arXiv, 2023.

Carion, Nicolas, et al. "SAM 3: Segment Anything with Concepts." ICLR, 2026.